

Lecture 13, 21.6.26

Exercise 1: prove or disprove: there exists a

fun. $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ for which

$$\nabla f(x,y) = \begin{pmatrix} x^2 y \\ y^2 x \end{pmatrix}, \quad \forall x,y \in \mathbb{R}$$

Solution: False. ATC that there exists such a fun.

then $f_x = x^2 y$, $f_y = y^2 x \quad \forall x,y \in \mathbb{R}$. It follows that

$$f_{xy} = x^2, \quad f_{yx} = y^2 \quad \forall x,y \in \mathbb{R}$$

Since f_{xy} and f_{yx} are cont. fun. by AOC,
then by theorem 1, lecture 12,

$$x^2 = f_{xy} = f_{yx} = y^2 \quad \forall x,y \in \mathbb{R}$$

contradiction.



Exercise 2: prove or disprove: there exists a

fun. $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ for which

$$\nabla f(x,y) = \begin{pmatrix} e^{x+y}(x+1) \\ x e^{x+y} \end{pmatrix} \quad \forall x,y \in \mathbb{R}$$

Solution: True. Choose $f(x,y) = x e^{x+y} \quad \forall x,y \in \mathbb{R}$.

Note that indeed

$$\nabla f(x,y) = \begin{pmatrix} f_x(x,y) \\ f_y(x,y) \end{pmatrix} = \begin{pmatrix} e^{x+y}(x+1) \\ x e^{x+y} \end{pmatrix}, \quad \forall x,y \in \mathbb{R}.$$

Exercise 3: prove or disprove: there exists a

fun. $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ for which

$$\nabla f(x,y) = \begin{pmatrix} y+2x \\ x \end{pmatrix} \quad \forall x,y \in \mathbb{R}$$

Solution: True. Choose $f(x,y) = xy + x^2, \quad \forall x,y$.

clearly, $f_x = y + 2x$ and $f_y = x$.



draft

$$\begin{aligned}
 f_x &= y + 2x \rightarrow f = xy + x^2 + C(y) \\
 f_y &= x \rightarrow f = \int x dy = xy + C(x) \\
 &\rightarrow f_x = y + C'(x) = y + 2x \\
 &\rightarrow C'(x) = 2x \rightarrow C(x) = x^2 + C \\
 &\xrightarrow{C=0} f = xy + C(x) = xy + x^2
 \end{aligned}$$

Theorem 1: let $\vec{u}, \vec{v} \in \mathbb{R}^n$ s.t. $\vec{u}, \vec{v} \neq \vec{0}$ and let $0 \leq \theta \leq \pi$ be the angle between $\vec{u}, \vec{v}$. Then

$$\langle \vec{u}, \vec{v} \rangle = \|\vec{u}\| \cdot \|\vec{v}\| \cdot \cos \theta$$

proof: we saw in linear algebra.



$$D(f)(\vec{u}) = a \cdot f_x + b \cdot f_y \quad \vec{u} = (a, b)$$

corollary 1: Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, and let $\vec{u} \in \mathbb{R}^2$ be a unit vector. If f_x, f_y are cont. at $\vec{x}_0$, then

$$\begin{aligned} D_{\vec{u}}(f)(\vec{x}_0) &= \langle \nabla f(\vec{x}_0), \vec{u} \rangle. \\ &= \|\nabla f(\vec{x}_0)\| \cdot \|\vec{u}\| \cdot \cos \theta \\ &= \|\nabla f(\vec{x}_0)\| \cdot \cos \theta. \end{aligned}$$

thus, $D_{\vec{u}}(f)(\vec{x}_0)$ is maximal when $\theta = 0$, i.e., maximal in the direction of the gradient, and minimal when $\theta = \pi$, i.e., in the opposite direction of the gradient. moreover, if $\theta = \frac{\pi}{2}$, then $D_{\vec{u}}(f)(\vec{x}_0) = 0$.

Exercise 4: consider the fun. $f(x,y) = x \ln y$.

prove or disprove: there exists a $\vec{u} \in \mathbb{R}^2$ s.t

$$D(f)(\vec{u})(1,1) = e$$

Solution: False. ATC that there exists such

a $\vec{u} \in \mathbb{R}^2$. Denote $\vec{v} = \frac{\vec{u}}{\|\vec{u}\|}$. Note $\vec{u} \neq \vec{0}$

(otherwise $D(f)(\vec{u})$ is not defined). thus $\vec{v}$

is well defined, and we know $\vec{v}$ is a unit vector. Denote $\vec{v} = (a,b)$. Then

$a^2 + b^2 = 1$. Note that $f_x = \ln y$

and $f_y = \frac{x}{y}$, so f_x, f_y are cont. at

$(1,1)$. By corollary 1,

$$\begin{aligned} D(f)(\vec{u})(1,1) &= D(f)(\vec{v})(1,1) = \\ &= \langle \nabla f(1,1), \vec{v} \rangle = \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \vec{v} \right\rangle = b = e \end{aligned}$$

thus $a^2 + e^2 = 1$. contradiction.



Exercise 5: consider the fun. $f(x,y) = x \ln y$.

Find a unit vector for which

$D(f)(1,1)$ is minimal

solution: we have $\vec{u} = \frac{-\nabla f(1,1)}{\|\nabla f(1,1)\|} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$.



Exercise 6: Find the line tangent to the ellipse $9x^2 + 16y^2 = 25$ at $(1,1)$.

solution: consider the fun.

$$f(x,y) = 9x^2 + 16y^2, \quad \forall x,y \in \mathbb{R}.$$

Then $9x^2 + 16y^2 = 25$ iff $f(x,y) = 25$.

We compute

$$\nabla f(x,y) = \begin{pmatrix} 18x \\ 32y \end{pmatrix}, \quad \forall x,y \in \mathbb{R}.$$

At $(1,1)$ we get

$$\nabla f(1,1) = \begin{pmatrix} 18 \\ 32 \end{pmatrix}.$$

We know the gradient is orthogonal to the direction of the tangent line.

We choose $\vec{u} = \begin{pmatrix} -32 \\ 18 \end{pmatrix}$, then $\langle \nabla f(1,1), \vec{u} \rangle = 0$

so the tangent line is in the direction of $\vec{u}$. thus the tangent line is

$\left\{ \begin{pmatrix} 1-32t \\ 1+18t \end{pmatrix} : t \in \mathbb{R} \right\}$. writing this line

in the form of an equation $y = ax + b$,

then

$$a = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 32 - 1}{19 - 1} = \frac{-32}{18}$$

plugging $(1,1)$ we get

$$1 = a \cdot 1 + b = \frac{-32}{18} + b$$

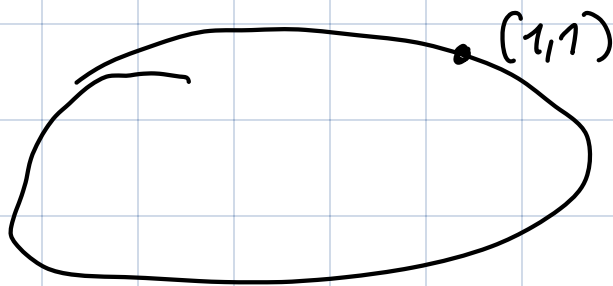
$$\text{thus } b = 1 + \frac{32}{18} = \frac{50}{18}$$

$$\Rightarrow y = \frac{-32}{18}x + \frac{50}{18}$$

Alternative solution:

$$9x^2 + 16y^2 = 25 \rightarrow 16y^2 = 25 - 9x^2$$

$$\rightarrow y = \pm \frac{1}{4} \sqrt{25 - 9x^2}$$



$$y(x) = \frac{1}{4} \sqrt{25 - 9x^2}$$

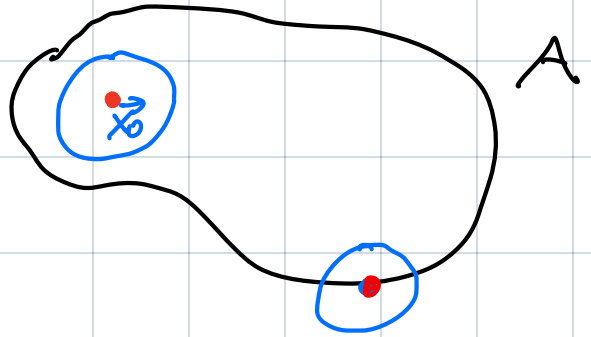
$$y'(x) = \frac{1}{4} \cdot \frac{1}{2\sqrt{25 - 9x^2}} \cdot (-18x)$$

$$a = y'(1) = \frac{1}{8 \cdot \sqrt{16}} \cdot (-18) = -\frac{18}{32}$$

$$1 = a \cdot 1 + b = -\frac{18}{32} + b$$

$$b = 1 + \frac{18}{32}$$

Definition: Let $A \subseteq \mathbb{R}^n$ and let $\vec{x}_0 \in A$. we say $\vec{x}_0$ is an interior point of A if $\exists r > 0$ s.t. $B(\vec{x}_0, r) \subseteq A$, where $B(\vec{x}_0, r)$ is the open ball of radius r centered at $\vec{x}_0$.



Definition: Let $A \subseteq \mathbb{R}^n$ and let $\vec{x}_0 \in \mathbb{R}^n$ (not necessarily in A). we say $\vec{x}_0$ is a boundary point of A if $\forall r > 0$, $B(\vec{x}_0, r) \cap A \neq \emptyset$ and $B(\vec{x}_0, r) \cap (\mathbb{R}^n \setminus A) \neq \emptyset$. The boundary of A is denoted by ∂A , and defined as the set of all boundary points of A .

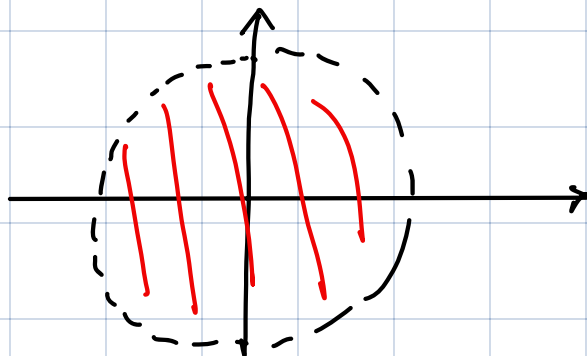
Definition: Let $A \subseteq \mathbb{R}^n$. we say A is closed if $\partial A \subseteq A$ and open if $\partial A \cap A = \emptyset$.

Example 5:

① the open unit disk

$$D = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x^2 + y^2 < 1 \right\}$$

is an open set. we have

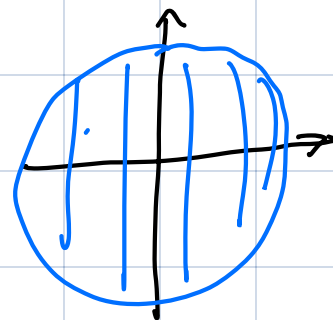


note that $\partial D = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : x^2 + y^2 = 1 \right\}$

② the closed unit disk

$$D = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x^2 + y^2 \leq 1 \right\}$$

is a closed set. we have

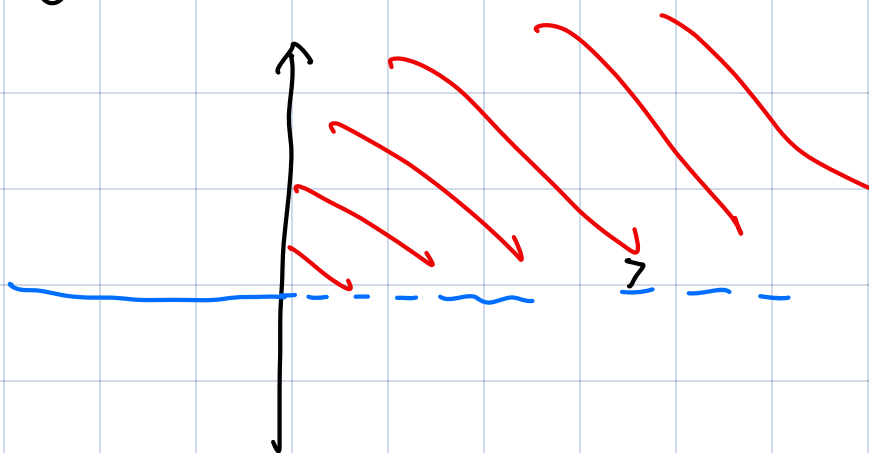


$$\partial D = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x^2 + y^2 = 1 \right\}$$

③ $\mathbb{R}^2$ is both open and closed.

④ $\emptyset$ is both open and closed.

⑤ $\left\{ \begin{pmatrix} x \\ y \end{pmatrix} : x \geq 0, y \geq 0 \right\}$



Definition: Let $f: D \rightarrow \mathbb{R}$ and let $\vec{x}_0 \in D$.

we say that

(i) $\vec{x}_0$ is a maximum of f in D if

$$f(\vec{x}_0) \geq f(\vec{x}), \quad \forall \vec{x} \in D$$

(ii) $\vec{x}_0$ is a local maximum of f in D if

$\vec{x}_0$ is an interior point in D and if $\exists r > 0$

s.t

$$f(\vec{x}_0) \geq f(\vec{x}), \quad \forall \vec{x} \in B(\vec{x}_0, r)$$

(iii) $\vec{x}_0$ is a minimum of f in D if

$$f(\vec{x}_0) \leq f(\vec{x}), \quad \forall \vec{x} \in D$$

(iv) $\vec{x}_0$ is a local minimum of f in D if

$\vec{x}_0$ is an interior point in D and if $\exists r > 0$

s.t

$$f(\vec{x}_0) \leq f(\vec{x}), \quad \forall \vec{x} \in B(\vec{x}_0, r)$$

Definition: Let $f: D \rightarrow \mathbb{R}$ and let $\vec{x}_0 \in D$. we

say $\vec{x}_0$ is a critical point of f if

(i) $\vec{x}_0$ is an interior point in D .

(ii) $\nabla f(\vec{x}_0) = \vec{0}$ or $\nabla f(\vec{x}_0)$ doesn't exist.

Theorem 2: (Fermat) let $f: D \rightarrow \mathbb{R}$ and let $\vec{x}_0 \in D$ be a local extremum. then $\vec{x}_0$ is a critical point.

Definition: Let $f: D \rightarrow \mathbb{R}$ and let $\vec{x}_0 \in D$ be a critical point. we say $\vec{x}_0$ is a saddle if $\vec{x}_0$ is not a local extremum.

Examples:

① consider the fun. $f(x,y) = x^2 + y^2, \forall x,y \in \mathbb{R}$.

Note $\nabla f(x,y) = \begin{pmatrix} 2x \\ 2y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. thus f
only one critical point, $(0,0)$. clearly,
 $(0,0)$ is a global minimum of f .

② consider the fun. $f(x,y) = -x^2 - y^2, \forall x,y \in \mathbb{R}$.

Note $\nabla f(x,y) = \begin{pmatrix} -2x \\ -2y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. thus f
only one critical point, $(0,0)$. clearly,
 $(0,0)$ is a global maximum of f .

③ consider the fun. $f(x,y) = x^2 - y^2, \forall x,y \in \mathbb{R}$.

Note $\nabla f(x,y) = \begin{pmatrix} 2x \\ -2y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. thus f
only one critical point, $(0,0)$. clearly,
 $(0,0)$ is a saddle point of f .

Theorem 3: (second derivative test) Let $D \subseteq \mathbb{R}^2$, let $f: D \rightarrow \mathbb{R}$ and let $\vec{x}_0 \in D$ be a critical point. Suppose all second derivatives are cont. at $\vec{x}_0$. Let H be the Hessian matrix of f at $\vec{x}_0$, namely,

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix}$$

Then

- (i) if $\det H > 0$ and $f_{xx} > 0$, then $\vec{x}_0$ is a local minimum.
- (ii) if $\det H > 0$ and $f_{xx} < 0$, then $\vec{x}_0$ is a local maximum.
- (iii) if $\det H < 0$ then $\vec{x}_0$ is a saddle

Exercise 7: consider the fun.

$$f(x, y) = x e^{-(x^2 + y^2)}, \quad \forall x, y \in \mathbb{R}.$$

- (a) Find all critical points of f .
- (b) classify each critical point as a local max, local min or a saddle

point.

